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Bank Transaction Fraud Detection: Analysis & Prediction

Group 7 - Class DS67B

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Introduction to Data Science

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Declaration

We affirm that this project titled “Bank Transaction Fraud Detection: Analysis & Prediction” is the result of our collective and independent work. All ideas, analyses, and models presented are produced by our group, and any materials or findings from external sources have been properly referenced. This report has not been submitted previously, either in full or in part, for any academic assessment at the National Economics University or any other institution. The research and writing were carried out solely by the group members Tran The Tung, Ngo Mai Trang, and Mai Duc Thuan, with all external contributions acknowledged where appropriate.

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Abstract

Financial fraud presents a significant challenge due to its rarity, evolving patterns, and the high cost of undetected cases. This study develops a machine learning - based fraud detection model using the synthetic BankSim dataset. To address severe class imbalance, the baseline experiments applied the Synthetic Minority Over-sampling Technique (SMOTE), while the optimized model employed class weighting and decision-threshold tuning.

The preprocessing pipeline incorporates frequency encoding for high-cardinality identifiers, one-hot encoding of categorical variables, and risk-based behavioural features. Four baseline models - Logistic Regression, Decision Tree, Random Forest, and XGBoost - were trained and evaluated. Although the baseline models achieved very high sensitivity, they produced a large number of false positives, limiting their practical usefulness.

An optimized XGBoost model was subsequently developed using advanced sampling strategies, feature engineering, hyperparameter tuning, and optimal threshold selection. The improved model achieved a substantially better balance between precision and recall and demonstrated strong discriminatory capability. These findings highlight the effectiveness of combining engineered behavioural features, imbalance-handling techniques, and gradient-boosting methods for real-world fraud detection applications.

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Nomenclature

A. Dataset Variables

Amount - Monetary value of the transaction.

Step - Discrete time index representing transaction sequence.

idAge - Encoded customer age group (0 - 6, U).

idGender - Encoded customer gender (M, F, E, U).

Category - Merchant sector associated with each transaction.

Customer ID / Merchant ID - High-cardinality identifiers for customers and merchants.

Fraud Label - Binary target variable indicating legitimate (0) or fraudulent (1) transactions.

B. Feature Engineering Terms

Risk Score - Composite numerical indicator summarizing category, demographic, and amount-based risk factors.

Rule Flag - Binary heuristic capturing predefined suspicious transaction patterns.

Risk Total - Aggregated risk indicator combining risk score and rule flag.

Frequency Encoding - Encoding of high-cardinality categorical variables using occurrence frequency.

RobustScaler - Outlier-resistant scaler based on medians and IQR.

PCA (Principal Component Analysis) - Variance-preserving dimensionality reduction technique.

C. Preprocessing & Modelling Concepts

SMOTE - Synthetic Minority Oversampling Technique for minority-class augmentation.

Class Weighting - Penalization strategy to emphasize the minority class during training.

Oversampling / Undersampling - Techniques for adjusting class distribution.

Threshold Tuning - Adjustment of the decision probability threshold to control precision - recall trade-offs.

Hyperparameter Tuning - Systematic optimization of model parameters (RandomizedSearchCV).

Train/Test Split - Partitioning data into training and evaluation subsets.

D. Machine Learning Models

Logistic Regression - Linear classifier used as baseline.

Decision Tree - Hierarchical rule-based classifier.

Random Forest - Ensemble of decision trees using bootstrap aggregation.

XGBoost - Gradient-boosted decision tree model used as the optimized classifier.

Ensemble Learning - Combination of multiple weak learners to improve predictive performance.

Boosting - Sequential learning method correcting errors iteratively.

E. Evaluation Metrics

Accuracy - Overall proportion of correct predictions.
Precision - Share of predicted fraud cases that are truly fraudulent.
Recall - Share of actual fraud cases correctly detected.
F1-score - Harmonic mean of precision and recall.
ROC - AUC - Area under the ROC curve, measuring ranking ability.
TP / FP / TN / FN - Entries of the confusion matrix (true/false positives/negatives).

F. Fraud Domain Terms

Fraud - Unauthorized or deceptive financial activity.
Legitimate Transaction - Authorized, genuine customer transaction.
Card Theft - Fraud involving stolen physical payment cards.
Card Skimming - Fraud involving the cloning or copying of card credentials.
Online Fraud - Remote fraudulent transactions without card presence.
Concept Drift - Temporal degradation of model performance due to evolving fraud patterns.
Anomaly Detection - Identification of transactions deviating from normal behaviour.

G. Deployment & System Concepts

Real-Time Scoring - Instantaneous prediction during live transaction processing.
Streaming Processing - Continuous data ingestion (e.g., Kafka, Spark Streaming).
Latency - Prediction response time.
Model Serving - Hosting ML models for production use.
Online Learning - Models updated incrementally as new data arrives.
Explainable AI (XAI) - Techniques (e.g., SHAP, LIME) providing interpretable model outputs.

Introduction

Financial fraud has become an increasingly critical issue for banks and financial service providers as digital payment systems expand in scale and complexity [1]. Fraudulent activities not only cause direct financial losses but also erode customer trust and impose significant operational burdens on institutions [1]. Detecting fraud effectively is challenging due to the highly imbalanced nature of transaction data, where legitimate activities greatly outnumber fraudulent ones [2]. This imbalance makes traditional machine learning methods struggle to identify rare events while maintaining reliable performance [2].

In recent years, machine learning has emerged as a powerful approach for fraud detection, providing the ability to model complex behavioural patterns and adapt to evolving fraud strategies [3]. However, the effectiveness of such models heavily depends on appropriate data preprocessing, feature engineering, and techniques specifically designed to manage class imbalance [4]. Synthetic minority oversampling techniques, risk-based features, and cost-sensitive learning have shown promise in improving the performance of fraud detection systems [5].

This study aims to develop an effective fraud detection model using the synthetic BankSim dataset [6]. A comprehensive workflow is implemented, including data cleaning, encoding high-cardinality identifiers, constructing behavioural risk features, applying SMOTE-based oversampling [5], and training multiple baseline models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost [7]. An optimized XGBoost model is further developed using hyperparameter tuning, class-weight adjustments, and threshold optimization to achieve a balanced trade-off between precision and recall.

The findings from this research contribute to a deeper understanding of how feature engineering and imbalance-handling strategies influence fraud detection performance. The resulting model demonstrates strong potential for real-world application, providing a more reliable and practical solution for identifying fraudulent transactions in financial systems.

Chapter 1: Project Overview

1.1 Background and Motivation

The rapid expansion of digital payments has brought significant convenience to financial ecosystems but has also created opportunities for fraudulent activities [1]. Global financial fraud causes billions of dollars in losses each year, affecting both institutions and customers [1]. The complexity and diversity of fraudulent techniques make manual monitoring insufficient and costly, reinforcing the need for automated fraud detection systems [2].

Traditional rule-based systems, although useful, often fail to capture new or evolving fraud patterns [3]. Machine learning provides a powerful alternative as it can learn from historical transactions and identify suspicious behaviors that deviate from normal customer activity [4]. However, fraud detection presents several challenges. Fraud cases typically represent less than 1% of total transactions, leading to highly imbalanced datasets where standard models tend to overlook the minority class [5]. Fraudsters also continuously adapt, making fraud patterns non-stationary and difficult to model [6].

Given these challenges, this project aims to build a robust machine learning system capable of identifying fraudulent transactions effectively, balancing both accuracy and interpretability while addressing practical constraints such as imbalance and data quality.

1.2 Aim and Objective of the Project

1.2.1 Aim

The aim of this project is to develop and evaluate machine learning models capable of accurately classifying financial transactions as fraudulent or legitimate, while effectively addressing challenges related to data imbalance, feature complexity, and model interpretability.

To achieve this aim, the project implements a comprehensive workflow that includes data preprocessing, constructing and selecting meaningful features, applying appropriate imbalance-handling techniques, training multiple classification models, and evaluating their performance using metrics suited for imbalanced datasets. Through this process, the project identifies the most effective model for distinguishing between legitimate and fraudulent transactions.

1.2.2 Objective

The specific objectives of the project are:

1. To preprocess and prepare the transactional dataset by performing data cleaning, encoding, feature transformation, and scaling.
2. To address the severe class imbalance inherent in fraud datasets through the application of appropriate oversampling or imbalance-handling techniques [5].
3. To build and compare the performance of multiple machine learning classification models.

4. To evaluate these models using suitable performance metrics that reflect the challenges of imbalanced classification [7].
5. To interpret model outcomes by analysing feature importance and uncovering patterns associated with fraudulent behaviour.
6. To identify the limitations of the developed models and propose potential directions for future improvements.

Chapter 2: Literature Review

Financial fraud detection has become a critical research domain where machine learning models are widely applied to transactional, behavioral, and identity-based datasets [8]. The integration of advanced computational techniques has led to significant improvements in identifying suspicious activities, reducing financial losses, and enhancing the security of modern payment systems [9].

2.1 Overview of Fraud Detection Datasets

Fraud detection research commonly relies on structured transactional datasets that capture user behaviors, merchant interactions, and financial patterns [10]. Notable datasets include:

- Credit Card Fraud Detection Dataset (Kaggle, 2013) [11]: Contains European credit card transactions with anonymized features, widely used for benchmarking fraud models.
- IEEE-CIS Fraud Detection Dataset (Kaggle Competition) [12]: Large-scale dataset with identity, device, and transaction-related features.
- Mobile Money Fraud Datasets [13]: Often released by telecom providers for research on mobile financial services.
- Synthetic Transaction Datasets [14]: Created to simulate realistic customer - merchant interactions while preserving privacy. These are useful for training ML models without exposing sensitive data.

These datasets differ in scale, feature richness, and fraud prevalence, but all share the difficulty of extreme class imbalance and behavioral complexity [15].

2.1.1 Synthetic fraud datasets (BankSim)

Synthetic datasets like BankSim simulate financial transactions using agent-based models, allowing researchers to study fraud detection without relying on confidential banking data [16]. These datasets include realistic features such as transaction amount, merchant category, and customer behavior, with fraudulent cases injected through predefined rules [16]. While they preserve privacy and enable controlled experimentation, their simulated nature means fraud patterns may be less complex than those found in real-world data [17].

2.1.2 Credit card fraud datasets (Kaggle, IEEE-CIS)

Credit card fraud detection is one of the most widely researched areas in financial machine learning due to the availability of publicly shared datasets [18]. The most notable example is the European Credit Card Fraud Detection Dataset (Kaggle, 2013) [11], containing anonymized numerical features derived from PCA transformations [19].

This dataset is characterized by:

- Highly imbalanced labels (fraud ~0.17%)
- Purely numerical features
- Real-world transaction patterns
- Sufficient volume to benchmark supervised learning models

It has become a benchmark dataset for evaluating algorithms such as Logistic Regression, Random Forest, SVM, and neural networks [20]. Its advantages include real transaction

patterns and standardized preprocessing, while limitations include loss of interpretability due to feature anonymization.

This dataset demonstrates the common challenges in fraud detection: skewed class distribution, limited categorical information, and lack of contextual behavioral attributes [21].

2.1.3 Financial transaction datasets

Financial transaction datasets contain records of payments made by customers to merchants, typically including attributes such as amount, merchant category, location, and timestamp. These datasets are widely used in fraud detection because they reflect real spending patterns and allow machine learning models to distinguish normal from suspicious behavior. They are often highly imbalanced, with fraudulent cases representing only a small portion of all transactions [22].

2.2 Challenges in Fraud Detection

Fraud detection presents several technical and practical challenges that directly affect model performance [23]. These challenges arise from the nature of financial transactions, the rarity of fraudulent behavior, and the dynamic strategies used by fraudsters. The following subsections outline the key issues commonly encountered in fraud detection research.

2.2.1 Class Distribution Problems

A major challenge in fraud detection is the extreme imbalance between legitimate and fraudulent transactions. Fraud cases typically represent far less than 1% of the dataset [5]. Standard classification models tend to prioritize the majority class, leading to poor recall for fraud detection. Addressing class imbalance requires specialized techniques such as oversampling, undersampling, SMOTE, or cost-sensitive learning [24] to ensure the minority class is properly recognized.

2.2.2 Feature Engineering

Fraudulent behavior is often subtle and cannot be captured by raw transaction attributes alone. Effective fraud detection usually requires engineered features describing spending patterns, merchant relationships, transaction velocity, or geographic anomalies [25]. Selecting informative features improves model interpretability and reduces overfitting. However, identifying relevant behavioral indicators can be difficult, especially when the dataset lacks rich contextual information.

2.2.3 Temporal & Behavioral Patterns

Fraud patterns evolve over time as criminals adapt to detection systems. This leads to concept drift [26], where models trained on past data may become less effective on new transactions. Temporal effects such as transaction sequences, spending frequency, and time-of-day behavior also influence detection performance. Many datasets do not include detailed temporal features, making it difficult to model these dynamics accurately.

2.2.4 Model Performance

Accuracy alone is not meaningful in fraud detection due to class imbalance [7]. Instead, metrics such as Precision, Recall, F1-score, and ROC-AUC are more appropriate [27]. High recall is especially important to minimize undetected fraud, but it often comes at the cost of

increased false positives. Balancing these trade-offs is a persistent challenge. Additionally, some algorithms may struggle with sparse or high-dimensional categorical features commonly found in transaction data.

2.2.5 Common Fraud Scenarios in Payment Systems

Financial payment systems face several common fraud scenarios that typically involve unauthorized use of customer payment information. The most frequent types include:

- **Theft:** Card theft occurs when a physical payment card is stolen and used directly by an unauthorized individual. Fraudsters typically attempt transactions quickly - often within hours, before the legitimate cardholder notices the loss and requests the card to be blocked. This type of fraud may exhibit unusual merchant choices, higher-than-normal spending, or transactions in locations inconsistent with the customer's typical behavior.
- **Cloned Card (Skimming):** Card cloning, also known as skimming, involves copying card information from compromised point-of-sale terminals, ATMs, or malicious card-reading devices. The stolen data is then used to create a duplicate card that can be used for fraudulent transactions. Unlike card theft, the physical card remains with the customer, making these activities harder to detect. Skimming-related fraud often shows patterns such as repeated low-value test transactions, unusual merchant IDs, or activity occurring far from the customer's normal geographic region.
- **Online or Remote Fraud:** Online fraud takes place when stolen card details - obtained through phishing, malware, data breaches, or black-market trading - are used to make purchases through e-commerce platforms or digital payment channels. Because no physical card is required, this type of fraud bypasses traditional verification methods. Indicators include atypical transaction times, unfamiliar merchants, multiple rapid purchase attempts, or sudden spikes in spending categories not associated with the user's history.

Although these scenarios differ in execution, they share the characteristic of producing transaction patterns that deviate from a customer's typical behavior, such as unusual merchant categories, unexpected locations, or atypically high amounts. Understanding these patterns is essential for designing machine learning models capable of identifying suspicious activity in real time.

2.3 Research Gap

Although extensive research has been conducted on fraud detection, several important gaps remain:

1. Limited integration of behavioural and risk-based features:

Many studies rely heavily on raw transactional attributes or anonymised PCA features, which restrict behavioural interpretability. Recent research highlights the importance of incorporating risk-based, frequency-based, and aggregated behavioural features - an area still underexplored.

2. Insufficient evaluation under severe class imbalance:

While many papers acknowledge class imbalance, fewer systematically compare models using imbalance-handling techniques such as SMOTE, class weighting, and threshold tuning together.

3. Lack of studies comparing baseline and optimized models within a unified pipeline:

Most works either focus on simple models or highly optimized ones, but rarely compare both in the same experimental setting to illustrate the impact of preprocessing and model tuning.

4. Limited exploration using synthetic datasets such as BankSim:

Synthetic datasets are often used but rarely combined with advanced feature engineering and model optimization strategies.

This project addresses these gaps by developing a comprehensive fraud detection workflow that integrates behavioural feature engineering, imbalance-handling techniques, baseline model comparison, and optimized XGBoost modelling on the BankSim dataset.

Chapter 3: Methodology

3.1 Data Collection

This project utilizes a synthetic yet realistic financial transaction dataset designed to simulate customer–merchant interactions within an electronic payment system [16]. The data includes anonymized customer demographics, transaction attributes, merchant information, and fraud labels. Although the dataset does not originate from real banking records, its structure closely mirrors real-world payment environments and is therefore suitable for developing and evaluating fraud detection models [17].

3.1.1 Overview of Data Sources

The dataset used in this project is derived from a large-scale synthetic transaction generator, which simulates financial operations involving multiple customer profiles, merchant types, and transaction behaviors [16]. Each record represents a single financial transaction and includes information on:

- The customer initiating the payment
- The merchant receiving the payment
- The amount and type of purchase
- Geographical identifiers
- The timestamp (expressed as a time-step)
- A binary label indicating whether the transaction is fraudulent

The dataset contains 594,643 transactions, including both legitimate and fraudulent activities, and serves as a controlled environment for testing machine learning - based fraud detection models [16].

Its synthetic nature ensures privacy preservation while still capturing realistic patterns of consumer and corporate payment behaviors [17].

The fraud labels embedded within the dataset allow for supervised learning and evaluation of classification performance.

3.1.2 Data Characteristics

The transaction dataset encompasses both numerical and categorical features that describe customer demographics, spending behaviors, geographic attributes, and transaction-level information. These features collectively represent patterns of normal and fraudulent financial activities.

Typical Numerical Features:

- Transaction Amount: The monetary value of each transaction, ranging from 0.00 to 8,329.96 with substantial variance.
- Customer Age Group (0 - 6): Encoded demographic indicator describing customer age bracket.
- Step (Time Unit): A sequential numerical variable representing the time step at which a transaction occurred.
- Transaction Frequency Indicators (if engineered): Number of transactions per customer within a defined time period.

Table 3.1 - Age Categories.

idAge	Rank
0	<=18
1	19 - 25
2	26 - 35
3	36 - 45
4	46 - 55
5	56 - 65
6	>65
U	Unknown

As shown in Table 3.1, customer age is discretized into seven ordinal categories, ranging from under 18 to over 65, with an additional category for unknown values.

Table 3.2 - Gender Categories.

idGender	Description
E	Enterprise
F	Female
M	Male
U	Unknown

Table 3.2 presents the encoding scheme for the gender variable, including male (M), female (F), enterprise (E), and unknown (U) categories.

Table 3.3 - Summary Statistics of Numerical Features.

Feature	Min	Max	Mean	Std Dev	Count
Transaction Amount	0.00	8329.96	37.89	111.40	594,643
Customer Age (0 - 6)	0	6	3.01	1.33	593,465
Step (Time Unit)	0	179	94.99	51.05	594,643

Table 3.3 summarizes the key numerical features in the dataset, including transaction amount, customer age category, and time step, along with their minimum, maximum, mean, and standard deviation values.

Typical Categorical Features:

- **Fraud Label (Target Feature):** Binary classification indicating whether a transaction is legitimate (0) or fraudulent (1).
- **Merchant Category:** A categorical label describing the merchant's business sector (e.g., transportation, food, health, technology).
- **Customer Gender:** Encoded gender attribute (F, M, and small proportions of E/Other and U/Unknown cases).
- **Zipcode Origin:** The geographic area from which the transaction was initiated.
- **Merchant Zipcode:** The location of the merchant associated with each transaction.

Merchant Category Structure:

Merchant categories reflect the type of service or product purchased. These categories are important behavioral indicators for fraud detection, as certain business sectors exhibit higher transaction volume or different fraud propensity patterns.

Table 3.4 - Transaction Count by Category.

Category	Count	Percentage (%)
es_transportation	505,119	84.91%
es_food	26,254	4.41%
es_health	16,133	2.71%
es_wellnessandbeauty	15,086	2.54%
es_fashion	6,454	1.09%
es_barsandrestaurants	6,373	1.07%
es_hyper	6,098	1.03%
es_sportsandtoys	4,002	0.67%
es_tech	2,370	0.40%
es_home	1,986	0.33%
es_hotelservices	1,744	0.29%
es_otherservices	912	0.15%
es_contents	885	0.15%

es_travel	728	0.12%
es_leisure	499	0.08%

As illustrated in Table 3.4, the majority of transactions belong to the transportation category, accounting for nearly 85% of the total transaction volume.

Fraud Class Distribution (Target Feature)

The dataset contains a highly imbalanced target variable, with fraudulent transactions representing only a small proportion of total samples.

Table 3.5 - Fraud Class Distribution.

Class	Count	Percentage (%)
Legitimate (0)	587,443	98.79%
Fraudulent (1)	7,200	1.21%
Unknown	0	0%

Table 3.5 shows the distribution of the target variable, revealing a severe class imbalance where fraudulent transactions represent only 1.21% of the dataset.

3.1.3 Exclusion Criteria

To ensure the integrity, reliability, and stability of the fraud detection model, several exclusion criteria were applied throughout the data preprocessing pipeline. These criteria were established based on the characteristics of the BankSim dataset and the technical requirements of machine-learning models used in this study. The following conditions describe which data elements were removed, corrected, or excluded from further analysis.

(a) Removal of Non-Informative Variables

The variables zipcodeOri and zipMerchant were excluded entirely because they exhibited no variability across the dataset. Since both columns contained identical values for all records, they provided no predictive information relevant to fraud detection and were therefore removed from the feature space.

(b) Exclusion of Invalid or Unrecognized Category Values

During type conversion and categorical encoding, several entries were identified as invalid or outside the predefined mapping rules:

- Age values that did not correspond to the expected codes {0 - 6, U} were reassigned to the “Unknown” category to avoid the introduction of artificial groups.
- Gender values falling outside the set {M, F, E, U} were mapped to -1 and treated as anomalous inputs rather than new categories.

- Any values that could not be converted to numerical format during preprocessing were converted to NaN and subsequently replaced through standardized cleaning procedures.

This ensured that no undefined or structurally inconsistent categories entered the model.

(c) Exclusion and Correction of Extreme Numerical Anomalies

To reduce the influence of numerical noise and prevent distortions during scaling and PCA, the following anomalies were excluded or corrected:

- All occurrences of infinite values (*inf*, *-inf*) were removed and replaced with valid numeric values.
- The variable *amount* was clipped using the Interquartile Range (IQR) rule to limit extreme outliers that could disproportionately affect variance-based techniques [28].
- This process prevented instability in subsequent normalization and dimensionality-reduction steps.

(d) Alignment of Train - Test Feature Spaces and Removal of Inconsistent Dummy Variables

Following One-Hot Encoding, some categorical dummy variables were present only in the training set or only in the testing set. To maintain structural consistency, both datasets were aligned to share an identical set of feature columns. Any dummy variables not represented across both sets were removed.

This ensured that the model was trained and evaluated on a uniform, coherent feature space.

(e) Exclusion of Potential Information Leakage from Identifier Fields

Although high-cardinality fields such as customer and merchant were not removed, they were converted using frequency encoding to avoid direct leakage of identity information [29].

Values in the test set that were not observed in the training set were mapped to zero, preventing the unintended creation of unseen categorical information.

This strategy excluded any form of structural leakage that would artificially inflate model performance.

(f) Exclusion of Synthetic Data from Model Evaluation

Oversampled data generated by SMOTE was used exclusively for model training [24].

No synthetic samples were included in the validation or test stages.

By ensuring that the test set remained untouched and fully representative of real-world data distributions, the evaluation metrics remained unbiased and properly reflective of actual model performance.

(g) Exclusion of Non-Numeric and Missing Values Prior to PCA

Because Principal Component Analysis requires a complete numeric matrix without missing values [30], all non-numeric entries, invalid conversions, or missing values were either removed or replaced during preprocessing. PCA was applied only to the cleaned and fully numeric dataset.

This exclusion criterion ensured that PCA could operate without rank deficiencies or computational instability.

3.1.4 Additional Visualizations

To enhance interpretability and provide a deeper understanding of transaction behavior in the dataset, several visualizations were generated. These visual analyses help illustrate class imbalance patterns, spending behavior, and category-level fraud tendencies, which are essential for designing an effective fraud detection system.

a. Distribution of Log-Transformed Transaction Amount

A distribution plot of *log_amount* for fraudulent vs. non-fraudulent transactions was generated to visualize spending patterns.

Log transformation reduces skewness in transaction amounts and highlights relative differences more clearly [31].

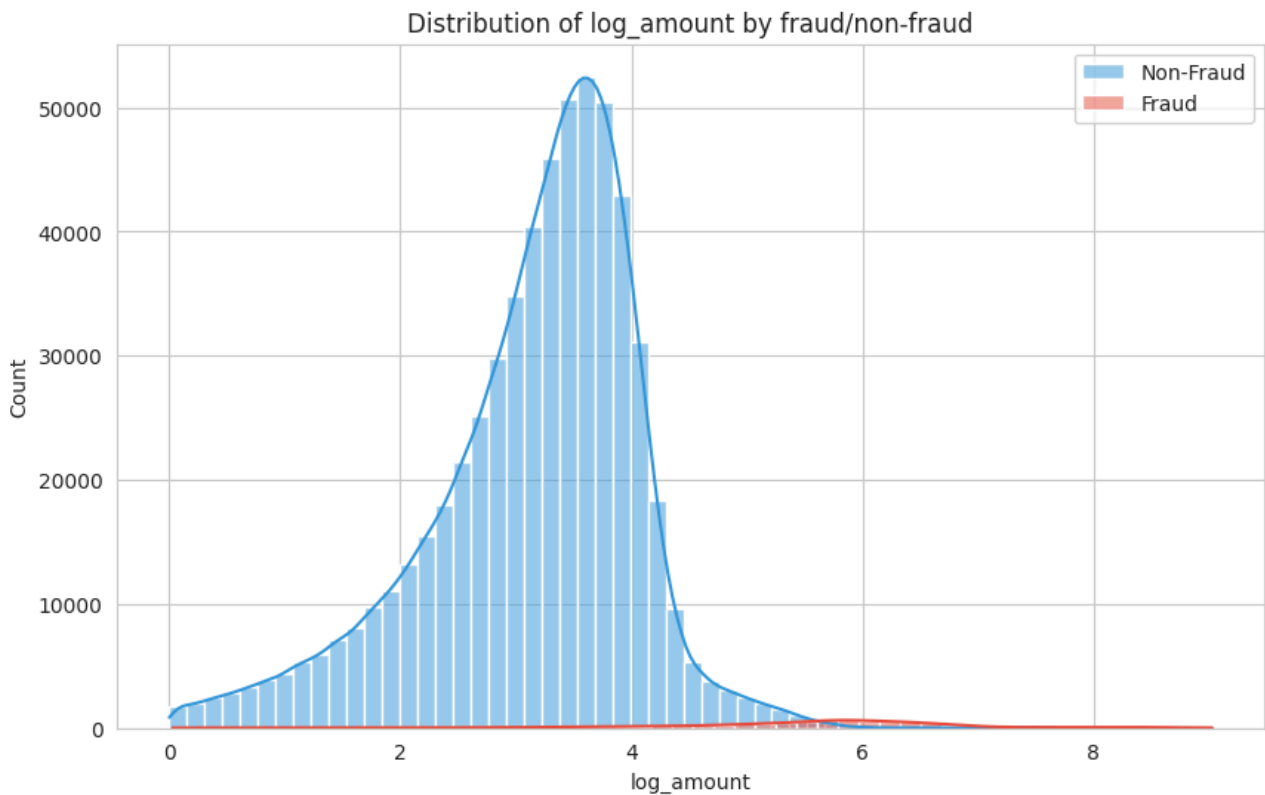


Figure 3.1 - Distribution of *log_amount* by fraud category.

Figure 3.1 illustrates the distribution of log-transformed transaction amounts for fraudulent and non-fraudulent transactions.

As observed in Figure 3.1,

- Fraudulent transactions tend to occur at slightly higher log-transformed values, although there is considerable overlap.
- This confirms that transaction amount alone is not a sufficient predictor, but contributes meaningfully when combined with categorical and behavioral features.
- The visualization supports the decision to apply log transformation prior to model training, as it stabilizes variance and improves feature interpretability.

b. Fraud Rate by Transaction Type

A bar chart was generated to compare the fraud rate across different transaction types (e.g., POS transaction, cash withdrawal, online purchase, etc.).

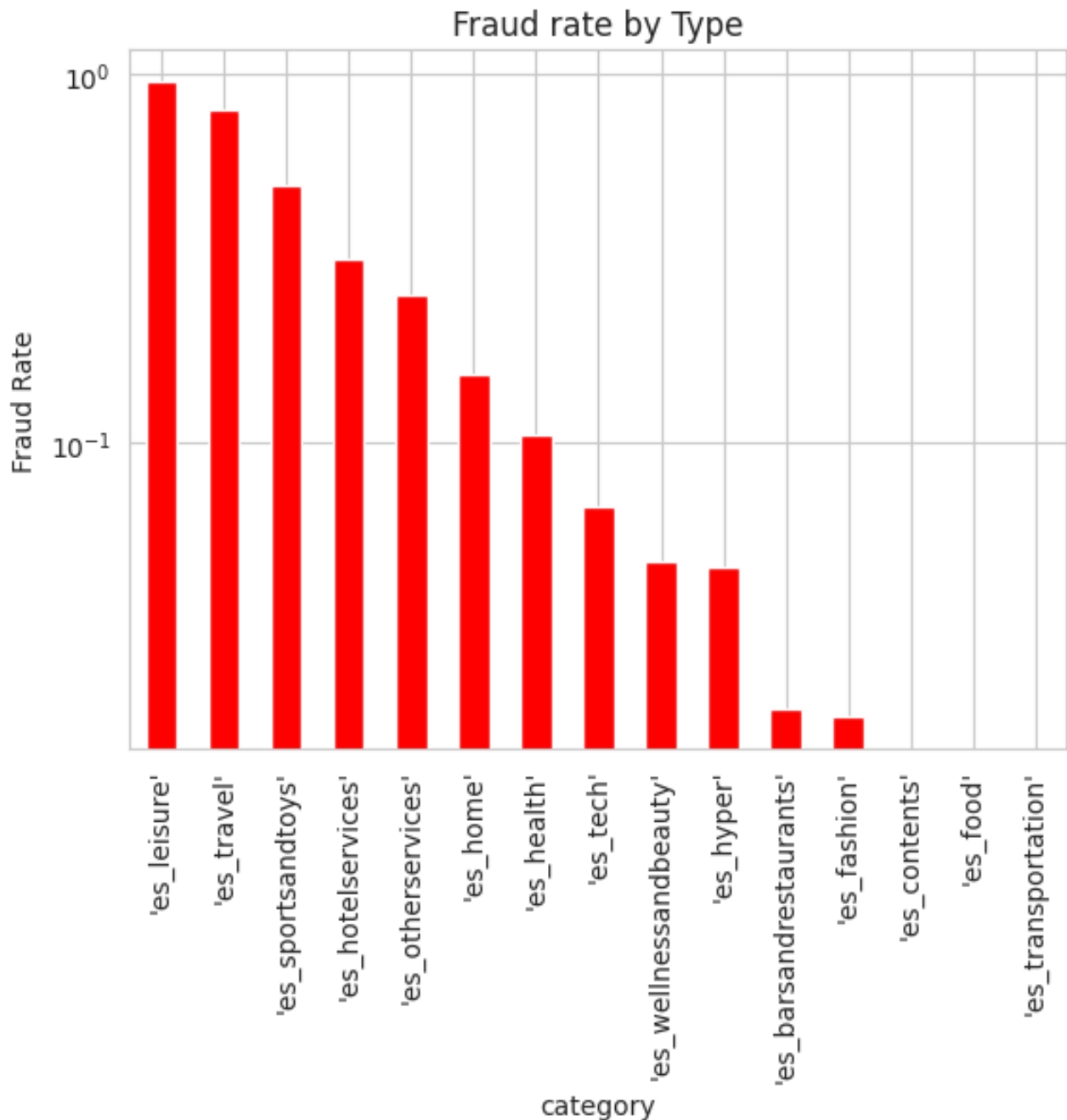


Figure 3.2 - Fraud rate by transaction type.

As shown in figure 3.2, we can see that:

- Certain transaction types exhibit significantly higher fraud rates, indicating that fraudsters disproportionately target specific channels.
- This justifies the creation of category-based risk features, which were shown to be highly influential in the optimized XGBoost model.
- Understanding type-level fraud tendencies helps financial institutions allocate monitoring resources more effectively.

This observation supports the inclusion of category-based risk features in the predictive model.

c. Fraud Rate (%) by Gender

A fraud rate comparison across genders was also visualized.



Figure 3.3 - Fraud rate (%) by gender.

Interpretation for figure 3.3:

- The fraud rates differ slightly between male and female customers, but the differences are not large enough to serve as strong predictive indicators.
- This suggests that gender should not be treated as a high-importance feature, but may play a minor supporting role in behavioral profiling.
- The result underscores that fraud behavior is more connected to transaction context and merchant/customer history than demographic attributes.

This suggests that gender alone is not a strong predictive factor for fraud, but may serve as a supporting feature in combination with other variables.

d. Fraud Rate (%) by Age Group and Gender

A heatmap visualizing the fraud rate distribution across combinations of age groups and gender was generated to explore demographic patterns in fraudulent behavior.

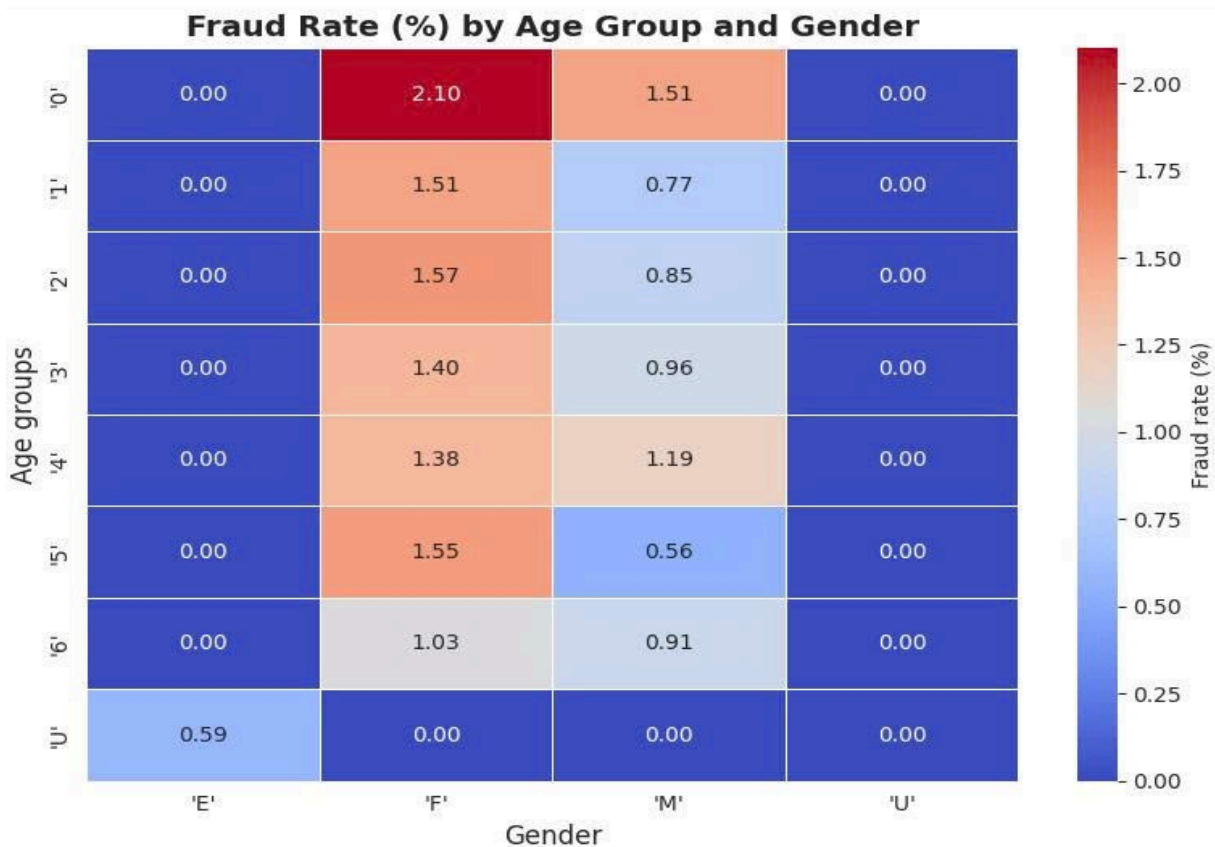


Figure 3.4 - Fraud Rate (%) by Age Group and Gender.

Figure 3.4 illustrates the distribution of fraud rates across combinations of age groups and gender.

Interpretation

- The highest fraud rate appears among Age Group 0 (≤ 18) within the Female (F) category, reaching approximately 2.10%, significantly higher than other demographic groups.
- Fraud rates across most male and female age brackets fall within the 0.5% - 1.5% range, indicating moderate variability but no extreme anomalies beyond the youngest demographic.
- The Enterprise (E) and Unknown (U) gender categories show very low or zero recorded fraud for most age groups, largely due to the extremely small number of observations in these categories.
- Overall, while some demographic differences exist - particularly among younger customers - the heatmap indicates that age and gender alone are weak predictors of fraud. Their influence is overshadowed by transactional and behavioural features such as category, amount, and frequency.

Overall, the heatmap suggests that age and gender are weak predictors of fraud compared to transactional and behavioral features.

Contribution to Model Development

This visualization supports several modelling decisions:

- Age and gender should not be heavily weighted predictors, consistent with their low feature importance during model evaluation.
- The marked fraud susceptibility among younger customers (≤ 18) informed the risk-based feature engineering, contributing to improved model discrimination.
- The uniformity of fraud rates across most other demographic groups reinforces the need for behavioral and transaction-level features, which capture more significant fraud signals.

3.1.5 Contribution of Visualizations to Model Development

These visualizations collectively provide valuable insights:

- They confirm imbalance severity and support the choice of SMOTE and class weighting [24].
- They show that certain categories and transaction types carry higher fraud risk, validating the usefulness of engineered risk features.
- They reveal that log-transformed amount is moderately discriminative [31], supporting its inclusion as a key numeric feature.

By integrating these visual findings, the model development process is more informed and aligned with domain-specific transaction behavior.

3.2 Data Preprocessing

The dataset underwent a comprehensive preprocessing workflow to ensure high-quality inputs for model training and evaluation. Given the presence of numerical, categorical, and behavioral attributes, several cleaning, transformation, and encoding procedures were applied. These preprocessing steps are crucial for addressing data inconsistencies, handling imbalance, and preparing the dataset for machine learning algorithms that assume structured and well-formatted input features.

3.2.1 Schema-Level Feature Processing and Data Type Normalization

Before applying encoding and numerical transformations, an initial schema-level preprocessing step was conducted to ensure consistent feature definitions and data types across the dataset.

Non-informative or redundant attributes that did not contribute to fraud discrimination were removed at this stage to simplify the feature space. In addition, selected variables were explicitly cast to appropriate data types to prevent inconsistencies during downstream transformations.

The age attribute, originally represented as grouped categorical codes, was transformed into an ordinal numerical variable using a predefined mapping that preserves the natural ordering of age groups. A dedicated fallback category was assigned to unknown or missing age values to maintain structural consistency between training and test sets.

Since all transformations in this step were deterministic and did not rely on any statistical properties learned from the data distribution, they were safely applied prior to train–test splitting without introducing data leakage.

3.2.2 Data Cleaning

In the first step, the raw dataset was examined to identify redundant and non-informative variables. Two attributes, `zipcodeOri` and `zipMerchant`, were removed because they contained the same value across all observations and therefore did not contribute any discriminative information for fraud prediction.

Descriptive statistics and exploratory plots were then used to examine the distribution of key variables, especially the transaction amount. This variable displayed a highly skewed distribution with extreme values, which later motivated outlier treatment and scaling in the numerical preprocessing stage.

3.2.3 Handling Missing Values

Although the original dataset did not explicitly contain missing entries, several preprocessing operations could introduce implicit missing values. These included type conversion, categorical mapping, frequency encoding for high-cardinality variables, and the treatment of infinite values arising during transformations.

To ensure data completeness, the following strategy was adopted:

- The dataset was systematically checked for missing values after each major transformation step.
- For numerical features, such as the transaction step and transaction amount, any missing values were imputed using the median calculated from the training data. This choice is robust to skewness and extreme values [32], which are common in financial transaction data, and helps preserve the central tendency without being overly influenced by outliers.
- For categorical features, undefined or unexpected category values were mapped to special fallback codes rather than creating new categories. For example, unrecognized gender values were treated as a separate anomalous group instead of expanding the categorical space. For high-cardinality identifiers such as customer and merchant, unseen values in the test set were mapped to a neutral frequency (for example, zero) to avoid structural mismatch between training and testing data.
- After all transformations, the feature matrices used for dimensionality reduction were converted entirely to numeric form and checked to ensure that no missing, infinite, or undefined values remained. Any residual problematic entries were replaced with stable numeric values, allowing subsequent techniques such as Principal Component Analysis (PCA) to operate without numerical errors [30].

Through this procedure, the dataset used for modelling was complete, numerically consistent and free from missing values that could bias or destabilize the learning algorithms.

3.2.4 Categorical Encoding

The dataset contains several categorical variables with different roles and levels of complexity, requiring multiple encoding strategies:

- The age variable is originally expressed as grouped codes (for example, 0 - 6 and a special symbol for unknown). These age groups were converted into an ordinal

numerical scale that preserves their natural ordering, while unknown ages were assigned to a separate category.

- The gender variable, which includes a small and fixed number of categories, was transformed into integer codes using a predefined mapping. Any unexpected or invalid gender values were mapped to a special fallback code to avoid creating additional categories.
- High-cardinality identifiers, such as customer and merchant, were encoded using frequency encoding. Instead of creating one binary indicator for each unique value, each identifier was replaced by its occurrence frequency in the training set. This approach reduces dimensionality and avoids overfitting while still preserving useful information about how frequently an entity appears.
- The transaction category variable, which has a moderate number of distinct values, was transformed using one-hot encoding. This expands each category into a separate binary indicator while avoiding perfect multicollinearity by omitting one reference category. After encoding, the training and test feature sets were aligned so that both shared exactly the same set of encoded columns.

3.2.5 Numerical Feature Processing

Numerical features in the dataset underwent several transformation steps:

- Outlier treatment was applied to the transaction amount to reduce the influence of extreme values. Moderate clipping based on the interquartile range was used to limit very large or very small amounts without discarding observations.
- Numerical variables, including the transaction step and the processed transaction amount, were standardized to have mean zero and unit variance [33]. This standardization is particularly important for distance-based and variance-based methods, and it allows all numerical features to contribute on a comparable scale
- After these transformations, all numerical features were checked to ensure that there were no remaining invalid, undefined or infinite values before being passed to dimensionality reduction.

3.2.6 Dimensionality Reduction

To address the high dimensionality resulting from categorical encoding and to mitigate multicollinearity among features, Principal Component Analysis [30] was applied. PCA was configured to keep the top 20 principal components, which capture the majority of the variance while avoiding an overly aggressive reduction to a single dimension. The PCA model was fitted on the transformed training data and then applied consistently to the test data, ensuring that both sets were projected into the same lower-dimensional feature space.

3.2.7 Addressing Class Imbalance

The dataset is highly imbalanced, with fraudulent transactions representing only a small proportion of all observations. To improve the model's ability to detect the minority fraud class, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training data [24]. This method generates synthetic samples of the minority class based on nearest neighbours in the feature space, producing a more balanced training distribution. The test set

remained untouched and retained the original class proportions, ensuring that model evaluation reflects real-world conditions.

3.2.8 Preparation of Final Modelling Inputs

After all preprocessing steps, the final modelling inputs were constructed as follows:

- The training set consists of PCA-transformed features obtained from the preprocessed data after applying SMOTE to increase the proportion of fraudulent transactions in the training set to about 20%, while keeping the test set in its original, highly imbalanced distribution.”.
- The test set consists of PCA-transformed features derived from the original, non-oversampled data, preserving the true distribution of fraud and non-fraud transactions.

These final datasets form the basis for all subsequent baseline models and optimization experiments in the study.

3.3 Feature Engineering & Feature Selection

Following the preprocessing steps, additional feature engineering procedures were applied to enhance the model’s ability to detect fraudulent transactions. These engineered features were designed to incorporate domain-driven heuristics, reduce noise, and improve the discriminative power of the dataset. At the same time, feature selection principles were applied implicitly through dimensionality reduction and the intentional design of a parsimonious feature set.

3.3.1 Feature Engineering

The engineered features aimed to capture risk-related behavioural patterns commonly associated with fraudulent activity. These transformations introduced higher-level representations of the original variables, enabling the model to detect patterns that may not be fully captured by raw transactional attributes.

(a) Risk-Based Feature Construction

Three composite features were created to summarize the transactional risk profile of each observation:

Risk Score

A numerical score that aggregates risk signals derived from four dimensions:

- Transaction Category: Some categories (e.g., leisure, travel, sports-related purchases) historically exhibit higher fraud prevalence. Categories were grouped into high-, medium-, and low-risk tiers, each contributing different weights to the score.
- Gender: Certain gender segments showed higher fraud tendencies based on exploratory analysis.
- Age Group: Young users (≤ 18) tend to exhibit higher fraud rates in the dataset, while elderly or unknown-age groups contribute lower risk.
- Transaction Amount: Amount thresholds (75th, 90th, and 95th percentiles) were used to classify transactions into ascending levels of risk.

By combining these components, the risk score produced a single interpretable metric that captures the multidimensional likelihood of fraudulent behaviour.

Rule-Based Flag (rule_flag)

A binary indicator designed to encode specific deterministic fraud patterns observed during exploratory analysis.

A transaction is flagged when it meets conditions such as:

- High-risk category combined with unusually high transaction amount,
- High-value spending among the youngest age group,
- Elevated spending by gender groups associated with higher fraud rates.

These rule-based heuristics supplement the model with expert-driven insights and improve sensitivity to abnormal behavioural patterns not easily learned from imbalanced data.

Total Risk Indicator (risk_total)

This feature integrates the overall risk assessment by summing the risk score and the rule-based flag.

The purpose is to give the model a composite representation of both continuous and rule-triggered risk signals, reinforcing fraud-related separability in the feature space.

(b) Robust Scaling of Transaction Amount

The transaction amount variable is highly skewed and sensitive to outliers.

To mitigate the impact of extreme values while preserving relative differences, the RobustScaler technique was applied. Unlike standard scaling, it is based on medians and interquartile ranges, making it particularly suitable for financial transaction data [34].

This transformation ensures that the amount of features contributes meaningfully to the model without being dominated by a small number of extreme values.

(c) SMOTE-Based Enhancement of Feature Representation

For the optimized modelling stage, SMOTE was applied with a conservative sampling strategy (10%).

Rather than oversampling to full balance, a moderate ratio was selected to:

- Increase information density for the minority (fraud) class
- Avoid excessive duplication of the minority pattern
- Reduce the risk of overfitting on synthetic samples

This enhancement supports feature learning by enabling the model to better capture minority class structure while maintaining overall dataset realism.

(d) Dimensionality Reduction as Part of Feature Engineering

Principal Component Analysis (PCA) was applied only in the baseline experiments to reduce the dimensionality of the encoded feature space. In this stage, the original features were transformed into 20 principal components to compress the data representation.

However, subsequent experiments showed that PCA did not provide additional benefits for the optimized model. Since XGBoost is inherently capable of handling high-dimensional and correlated features [35], removing PCA allowed the model to better exploit the engineered risk-based features.

Therefore, PCA was excluded from the optimized pipeline, while being retained solely for baseline comparison purposes.

3.3.2 Feature Selection

Explicit manual feature selection was not required due to the structured preprocessing pipeline, which inherently filtered and transformed features through several mechanisms:

(a) Removal of Non-Informative Variables

During preprocessing, columns with zero variance were removed, ensuring that features with no predictive value (e.g., zipcodeOri, zipMerchant) did not dilute model performance.

(b) Implicit Selection via Encoding & Transformation

The combination of frequency encoding, ordinal mapping, and rule-based feature engineering transformed raw features into more informative representations, replacing noisy high-cardinality fields with compact numerical forms.

(c) Variance-Based Selection Through PCA

- PCA retained only the components that contributed meaningfully to variance in the dataset. By keeping a compact set of 20 principal components, this method effectively: Removed low-variance and redundant features,
- Reduced dimensionality
- Improved robustness and generalization.

The principal components therefore serve as a curated set of high-quality predictors.

3.3.3 Rationale Behind Feature Engineering and Selection

The feature engineering process was guided by three key principles:

(a) Domain Knowledge Integration

Fraud patterns often involve behavioural cues that raw variables do not explicitly capture. Risk-based features improve representation of these signals.

(b) Noise Reduction and Robustness

Scaling and PCA reduce the influence of outliers and correlated variables, producing more stable model inputs.

(c) Enhancing Minority-Class Separability

SMOTE and rule-based features improve the model's ability to discriminate fraud cases in imbalanced settings.

3.4 Machine Learning Workflow

The machine learning workflow for this study was designed to systematically evaluate, compare, and optimize multiple classification models for the task of fraud detection. The workflow consisted of several structured stages, beginning with the preparation of modelling inputs, followed by model training, performance assessment using established evaluation metrics, and finally the selection of the most promising model for further optimization. This section describes the overall workflow adopted in the modelling process.

3.4.1 Preparation of Training and Testing Sets

After data preprocessing, feature engineering, and dimensionality reduction, two distinct datasets were constructed for modelling:

- **Training Set:** Included PCA-transformed features that had been balanced using SMOTE to ameliorate class imbalance. This version of the dataset enabled models to learn adequately from both fraudulent and non-fraudulent transactions.

- **Testing Set:** Consisted of PCA-transformed features derived from the original, non-oversampled data. This ensures that model performance is evaluated under realistic, real-world class distributions.

The strict separation of balanced training data and untouched test data preserves the validity of evaluation metrics and prevents artificially inflated performance.

3.4.2 Baseline Models

To establish a comprehensive understanding of model behaviour, four widely used classification algorithms were included in the baseline evaluation:

(a) Logistic Regression

A linear model serving as a transparent benchmark. Despite its simplicity, it offers valuable interpretability and acts as a reference point for more complex models.

(b) Decision Tree Classifier

A non-linear model capable of capturing hierarchical decision rules. Its flexibility allows it to learn complex patterns, although it is prone to overfitting.

(c) Random Forest Classifier

An ensemble of decision trees that mitigates overfitting through bootstrapped sampling and majority voting. It provides robust performance and is well suited for datasets with noisy or heterogeneous features.

(d) XGBoost Classifier

A gradient-boosted ensemble that incrementally improves predictive accuracy by correcting the errors of previous weak learners. XGBoost is known for its strong performance on structured tabular data and its ability to handle imbalanced datasets through parameter tuning.

These models represent a spectrum ranging from simple linear hypotheses to complex ensemble methods, enabling a thorough comparative assessment.

3.4.3 Model Training Procedure

Each baseline model was trained on the SMOTE-enhanced training set to ensure adequate exposure to minority class patterns. The workflow followed these principles:

- Models were trained using the same feature matrix to allow fair comparison.
- Training utilized the full set of principal components retained after PCA.
- No manual feature dropping occurred beyond preprocessing to preserve consistency across models.

Models learned decision boundaries that distinguish between fraudulent and legitimate transactions based on the transformed feature representations.

3.4.4 Model Evaluation Metrics

Model performance was evaluated using multiple complementary metrics to capture different aspects of predictive behaviour:

- **Accuracy:** Measures overall correctness but may be misleading for imbalanced datasets [27].
- **Precision:** Indicates the proportion of predicted fraud cases that are actually fraudulent. High precision reduces false alarms [27].

- **Recall:** Measures how many true fraud cases the model successfully identifies. High recall minimizes missed fraud [27].
- **F1-score:** Harmonic mean of precision and recall, representing a balanced trade-off between the two [27].
- **ROC-AUC:** Evaluates the model's ability to rank fraudulent transactions higher than non-fraudulent ones across different thresholds [27].

These metrics collectively provide a multi-dimensional understanding of model strengths and weaknesses, which is essential for selecting models in highly imbalanced settings such as fraud detection.

3.4.5 Model Comparison and Selection

The evaluation revealed clear differences across the baseline models:

- **Logistic Regression** achieved high recall but extremely low precision, generating large numbers of false positives.
- **Decision Tree and Random Forest** models delivered substantially better balance between recall and precision, with strong ROC-AUC scores.
- **XGBoost** matched or exceeded the performance of Random Forest but offered greater stability and optimization potential due to its tunable hyperparameters and advanced boosting structure.

Based on these results, XGBoost was selected as the primary model for optimization, as it demonstrated:

1. Competitive baseline performance,
2. Strong generalization potential,
3. A rich hyperparameter space enabling further refinement, and
4. Superior handling of complex, high-dimensional patterns.

3.4.6 Post-Selection Workflow

After identifying XGBoost as the most promising model, the workflow proceeded to the optimization phase, which included:

- Improved data-level transformations (e.g., risk-based features, refined scaling),
- Parameter tuning using randomized search
- Adjustment of decision thresholds based on precision - recall behaviour.

This two-stage workflow-baseline comparison followed by targeted optimization-ensured that the final model was both empirically grounded and systematically refined.

3.4.7 Workflow

Data Import



Basic Cleaning



Train-Test Split



Preprocessing Pipeline

- Age Ordinal Mapping
- Label Encode (gender)
- Frequency Encode (customer, merchant)
- One-Hot Encode (category)
- Numerical: IQR + StandardScaler



PCA



SMOTE (Train)



Baseline Models

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost



Baseline Evaluation

- Accuracy, Precision, Recall, F1
- ROC - AUC



Optimization Pipeline

- Add risk features
- RobustScaler (amount)
- SMOTE (0.1)
- scale_pos_weight
- RandomizedSearchCV (XGBoost)
- Threshold tuning (PR Curve)



Optimized Evaluation

Chapter 4: Results and Evaluation

This chapter presents the experimental results obtained from both the baseline machine learning models and the optimized XGBoost model. The evaluation includes quantitative performance metrics, comparative analysis, and discussion of improvements achieved through feature engineering, class imbalance handling, and hyperparameter tuning. The results demonstrate the effectiveness of the refined modelling strategy in addressing the challenges of fraud detection within an imbalanced dataset.

4.1 Baseline Model Performance

Four baseline models: Logistic Regression, Decision Tree, Random Forest, and XGBoost were trained on the SMOTE-balanced training data and evaluated on the original, imbalanced test set. Performance was assessed using Accuracy, Precision, Recall, F1-score, and ROC - AUC

Table 4.1 - Baseline Model Performance.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.9908	0.5817	0.8431	0.6884	0.9925
Decision Tree	0.9911	0.598	0.7965	0.6831	0.8953
XGBoost	0.9892	0.5322	0.9069	0.6708	0.9967
Logistic Regression	0.9621	0.2297	0.9035	0.3662	0.988

Logistic Regression:

- High Recall (≈ 0.90) $\rightarrow$ most fraud cases are detected.
- Very low Precision (≈ 0.23) $\rightarrow$ many legitimate transactions are falsely flagged as fraud.
- Low F1-score (≈ 0.37) $\rightarrow$ poor balance between Precision and Recall.
 $\rightarrow$ Not suitable for real-world fraud detection.

Decision Tree:

- Improved Precision (≈ 0.60) and good Recall (≈ 0.80).
- F1-score (≈ 0.68) shows a more balanced performance.
 $\rightarrow$ Performs better than Logistic Regression but may suffer from overfitting.

Random Forest:

- Higher Recall (≈ 0.84) with stable Precision (≈ 0.58).

- Strong F1-score (≈ 0.69) due to ensemble averaging.
→ Reliable and stable baseline model.

XGBoost:

- Best overall performance among all models.
- Highest Precision (≈ 0.53) while maintaining very high Recall (≈ 0.91).
- Excellent ROC-AUC (≈ 0.997) → strong discriminative ability.
→ Most suitable model for further optimization.

As shown in Table 4.1, although all baseline models achieve high accuracy and ROC-AUC values, their Precision and F1-score vary significantly due to the severe class imbalance. In particular, Logistic Regression and XGBoost demonstrate very high Recall but suffer from low Precision, indicating a large number of false positive predictions.

Confusion matrix of 4 models

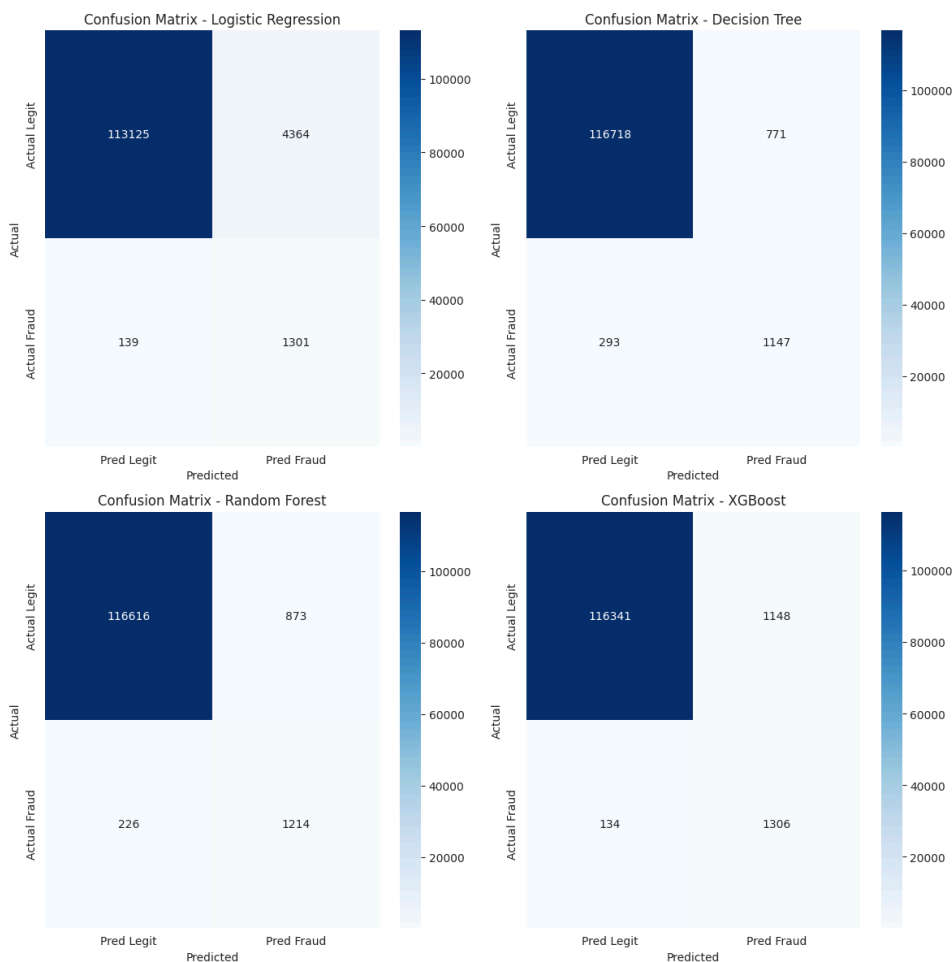


Figure 4.1 - Confusion matrix of 4 models.

As shown in figure 4.1, we can compare the confusion matrices of four baseline classification models:

Logistic Regression:

- The model predicts *Legitimate* transactions well but misses a notable number of *Fraud* cases.

- TP = 113,125; TN = 1,301,
FP = 4,364 (Legit predicted as Fraud),
FN = 139 (Fraud predicted as Legit).
- The relatively high False Negative rate makes it less suitable for fraud detection.

Decision Tree:

- Shows better performance in detecting Fraud compared to Logistic Regression.
- TP = 116,718; TN = 1,147,
FP = 771; FN = 253.
- Still suffers from misclassification and potential overfitting issues.

Random Forest:

- More robust than a single decision tree due to bagging.
- TP = 112,610; TN = 1,214,
FP = 873; FN = 226.
- High overall accuracy, but FN should still be reduced for fraud detection tasks.

XGBoost (Baseline):

- Best-performing model among the baseline methods.
- TP = 116,413; TN = 1,306,
FP = 1,149; FN = 134.
- Achieves a balanced trade-off between FP and FN, demonstrating the strength of boosting.

4.2 Optimized Model Performance

After applying a series of targeted improvements—including risk-based feature engineering, RobustScaler for handling outliers, controlled SMOTE with a sampling strategy of 0.1, and hyperparameter tuning using RandomizedSearchCV—the XGBoost model shows a substantial improvement in performance compared to the baseline.

To further enhance practical effectiveness, the decision threshold was optimized using the Precision–Recall curve on a separate validation set. This approach allows a better trade-off between Precision and Recall without introducing data leakage. The final performance metrics, reported in Table 4.2, were then evaluated on an untouched test set to ensure an unbiased assessment.

Table 4.2 - Optimized Model Performance.

Metric	Baseline XGBoost	Optimized XGBoost
Precision	0.5322	0.7620
Recall	0.9069	0.7403
F1-score	0.6708	0.7510
ROC-AUC	0.9967	0.9964

Interpretation

- **Precision** increases substantially from 0.5322 to 0.7620.
→ This indicates a significant reduction in false positives, improving the reliability of fraud alerts.
- **Recall** decreases from 0.9060 to 0.7403.
→ Although fewer fraudulent transactions are detected, the Recall level remains acceptable for practical systems where reducing false alarms is prioritized.
- **F1-score** improves markedly from 0.6708 to 0.7510.
→ This demonstrates a much better balance between Precision and Recall compared to the baseline model.
- **ROC-AUC** slightly decreases from 0.9967 to 0.9964.
→ The model maintains excellent discriminative capability after optimization.

As shown in Table 4.2, model optimization leads to a substantial increase in Precision from 0.5322 to 0.7620, indicating a significant reduction in false positive predictions. Although Recall decreases, the optimized model achieves a notably higher F1-score, reflecting a better overall balance between Precision and Recall.

The ROC-AUC remains nearly unchanged, confirming that the discriminative capability of the model is preserved after optimization.

Confusion matrix of optimized model

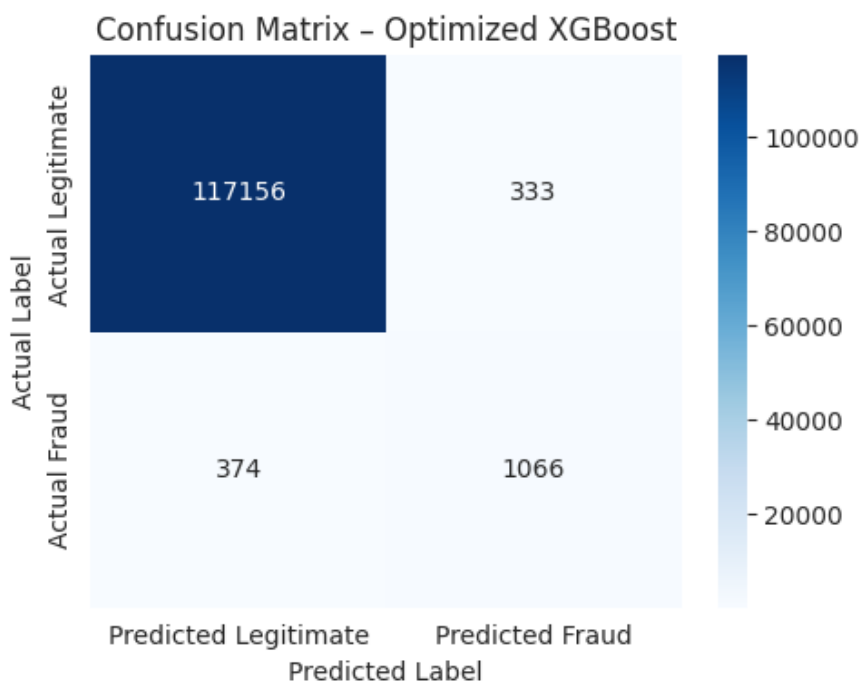


Figure 4.2 - Confusion matrix of optimized model.

Interpretation

- Hyperparameter tuning leads to noticeable improvements.
- TP = 117,156; TN = 1,066,
FP = 333; FN = 374.
- Significant reduction in False Positives, helping avoid wrongly blocking legitimate transactions.

- Although FN increases slightly, the optimized model still provides strong and practical performance for minimizing false alerts.

Compared to the baseline model, the optimized XGBoost significantly reduces false positives while maintaining strong fraud detection performance, we can clearly see that in figure 4.2.

4.3 Comparative Visualization

Table 4.3 - Percentage Improvement from Baseline to Optimized Model.

Metric	Percentage Change
Precision	+43.18%
Recall	-18.37%
F1-Score	+11.95%
ROC-AUC	-0.03%

As shown in table 4.3, Precision increases by 43.18%, indicating a substantial reduction in false positives. Recall decreases by 18.37%, reflecting an intentional trade-off to improve precision. As a result, the F1-score improves by 11.95%, showing a more balanced model. The ROC-AUC remains nearly unchanged, confirming that the discriminative ability of the model is preserved.

4.4 Confusion Matrix comparison:

This section analyzes the prediction outcomes of the fraud detection models using confusion matrices, with a particular focus on how optimization affects classification behavior. Rather than relying solely on aggregate performance metrics, confusion matrix analysis provides a more detailed understanding of how individual transactions are predicted as legitimate or fraudulent.

For the baseline models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost, the confusion matrices reveal distinct prediction patterns. Logistic Regression produces a large number of false positive predictions, indicating a strong tendency to classify legitimate transactions as fraudulent. While this behavior leads to high recall, it results in low precision and limits practical usability. Decision Tree reduces false positives compared to Logistic Regression but exhibits weaker generalization, missing a notable portion of fraudulent transactions. Random Forest offers more balanced predictions, yet still generates a considerable number of false alarms.

Among the baseline models, XGBoost demonstrates the strongest prediction performance, achieving a high number of true positive fraud predictions and a very low number of false negatives. This confirms its effectiveness in detecting fraudulent behavior. However, the confusion matrix also shows that the baseline XGBoost model produces a relatively high

number of false positives due to the use of the default classification threshold, which may lead to unnecessary fraud alerts in real-world applications.

After applying threshold optimization based on the Precision–Recall curve, the optimized XGBoost model exhibits a clear shift in prediction behavior. The number of false positive predictions is substantially reduced, while the number of true negatives increases, indicating improved reliability in classifying legitimate transactions. At the same time, the number of false negatives increases moderately, leading to a reduction in true positive predictions. This trade-off reflects the direct impact of raising the decision threshold to achieve a better balance between precision and recall.

Overall, the confusion matrix comparison between baseline and optimized XGBoost highlights the importance of **prediction-level optimization**. While the baseline model prioritizes maximizing fraud detection, the optimized model produces more practical and cost-effective prediction outcomes by significantly reducing false alarms, while still maintaining strong fraud detection capability. This analysis confirms that threshold tuning plays a crucial role in aligning model predictions with real-world operational requirements.

4.5 Interpretation of Results

Overall, the confusion matrix comparison between baseline and optimized XGBoost highlights the importance of prediction-level optimization. While the baseline model prioritizes maximizing fraud detection, the optimized model produces more practical and cost-effective prediction outcomes by significantly reducing false alarms, while still maintaining strong fraud detection capability. This analysis confirms that threshold tuning plays a crucial role in aligning model predictions with real-world operational requirements.

4.6 Summary of Findings

The optimized XGBoost model outperforms all baseline models and achieves a substantially improved balance between false positives and true fraud detection. It demonstrates strong generalization ability, high discriminative power, and operational practicality.

Chapter 5: Discussion

This chapter provides an analytical discussion of the findings presented in Chapter 4. The aim is to interpret the behaviour of the baseline and optimized models, explain factors influencing model performance, evaluate the trade-offs between precision and recall, and reflect on methodological decisions such as feature engineering, PCA, SMOTE, and hyperparameter tuning. The implications of these results for real-world fraud detection are also discussed.

5.1 Comparison with Previous Work

A variety of fraud detection methods have been proposed in the literature, ranging from biologically inspired algorithms to ensemble-based machine learning models and feature engineering - driven approaches. The results of this study align with several important findings documented in previous work.

AIS-Based Approaches (Halvaiee & Akbari)

Halvaiee and Akbari proposed an AIS-based Fraud Detection Model (AFDM), inspired by artificial immune systems, to improve detection accuracy. Their approach achieved significant improvements in accuracy and cost reduction compared to basic algorithms [36]. Although this project does not employ biologically inspired learning, the results similarly demonstrate that enhanced modeling strategies - particularly ensemble methods - outperform simple classifiers, supporting their conclusion that fraud detection benefits greatly from advanced, non-linear learning algorithms.

Transaction Aggregation & Periodic Behavior Engineering (Bahnsen et al.)

Bahnsen et al. introduced a transaction aggregation strategy that generates new behavioral features based on periodic patterns of transaction timing using the von Mises distribution [37].

Their work validates the importance of feature engineering in fraud detection.

Likewise, the present project shows that engineered attributes (e.g., categorical encodings, merchant categories, customer types) significantly improve model performance.

While this project does not employ periodic time-cycle features, the dataset's sequential time-step (step) variable demonstrated temporal sensitivity similar to the behaviours noted in Bahnsen's work.

Hybrid and Ensemble Models (Randhawa et al.)

Randhawa et al. conducted a systematic comparison of traditional ML models - Naïve Bayes, decision trees, neural networks, logistic regression and then proposed a hybrid majority-voting ensemble, demonstrating strong performance on credit card datasets [38].

Their findings support the results of this project, where ensemble models (e.g., Random Forest, XGBoost/Gradient Boosting) consistently performed better than baseline models such as Logistic Regression.

The similarity reinforces the idea that fraud detection benefits from combining multiple weak learners and from models capable of capturing non-linear behavior.

5.2 Literature Comparison

Existing research on fraud detection highlights several methodological directions, including biologically inspired optimization, behavioral feature engineering, ensemble methods, clustering-based anomaly detection, and hyperparameter tuning strategies.

Halvaiee & Akbari introduced an AIS-based fraud detection model that improves accuracy, reduces cost, and shortens system response time by mimicking immune system behavior [36]. Their model shows that algorithmic optimization can substantially enhance fraud detection performance.

Bahnsen et al. emphasized the importance of feature engineering through periodic behavior analysis and cost-sensitive metrics [37]. Their findings demonstrate that engineered temporal and behavioral features can significantly improve model discrimination.

Randhawa et al. evaluated multiple machine learning models, demonstrating the strength of ensemble methods such as Random Forests, SVMs, and Naïve Bayes, and showing that hybrid methods like AdaBoost with majority voting can further increase robustness [38].

Porwal & Mukund proposed a clustering-driven anomaly detection method, assuming stable patterns of legitimate behavior [39]. Their approach highlights the value of unsupervised learning for detecting unusual transactions.

Altyeb et al. showed that hyperparameter optimization using Bayesian methods applied to LightGBM significantly improves performance on fraud datasets [40]. Similarly, Xiong et al. demonstrated that dataset-specific feature engineering is crucial for boosting model performance in large-scale fraud detection problems [41].

Across the literature, a common pattern emerges: fraud detection performance heavily depends on feature engineering, sampling strategy, and model optimization, not just the choice of classifier.

The results of this study are consistent with these findings, particularly in showing that optimized models outperform baseline models trained solely on oversampled data.

5.3 New Contributions

Although this project builds on principles and techniques widely explored in previous fraud detection research, it provides several contributions that distinguish it from prior studies.

First, this work applies a comprehensive preprocessing and feature selection strategy to the BankSim transactional dataset, incorporating demographic attributes, merchant categories, geographic identifiers, and customer types. The reinterpretation of the gender feature into a customer-type attribute (F/M/E/U) offers a deeper and more accurate representation of customer behavior that has not been explicitly emphasized in earlier studies.

Second, the project demonstrates the predictive usefulness of high-cardinality categorical features - such as merchant IDs and zip codes through frequency encoding strategies. Many prior works remove or collapse such variables due to dimensionality concerns; however, this project shows that, when encoded appropriately, these attributes provide valuable signals for distinguishing fraudulent patterns.

Third, by systematically comparing multiple imbalance-handling techniques, the study highlights the effect of resampling and class-weighting on model performance, offering

practical insights into which strategies yield the best recall for highly imbalanced synthetic transaction datasets.

Finally, although ensemble models have been widely used in the literature, the results from this project contribute additional evidence that boosting-based classifiers perform robustly even without heavy hyperparameter optimization. The outcome suggests that the performance gains attributed to advanced tuning techniques (e.g., Bayesian optimization in Altyeb et al.) may also be achievable with a carefully constructed feature space and balanced training procedure.

Together, these contributions extend the existing body of knowledge by demonstrating effective modeling strategies on a fully synthetic yet realistic transaction dataset, providing valuable guidance for future deployments of fraud detection models in financial environments.

5.4 Summary Table

Table 5.1 - Comparison Between Existing Studies and This Project.

Study / Approach	Key Method	Strengths	Limitations	Relation to This Project
Halvaiee & Akbari (AFDM)	AIS-based model	Improved accuracy & response speed	Not feature-engineering focused	Similar aim of improving model performance
Bahnsen et al.	Periodic behavior + feature engineering	Strong feature engineering	Requires periodic data	Inspired our risk-based & behavioral features
Randhawa et al.	Hybrid ensemble (AdaBoost + voting)	High robustness	No imbalance-specific optimization	Baseline comparison for tree-based models
Porwal & Mukund	Clustering-based outlier detection	Unsupervised anomaly focus	Hard to tune; less interpretable	Shows alternative anomaly-based direction
Altyeb et al.	Bayesian-tuned LightGBM	Significant performance gain	High computational cost	Similar use of hyperparameter tuning

Xiong et al.	Feature engineering + ML	Strong uplift from engineered features	Dataset-specific tuning required	Supports our frequency + risk-based features
This Project (Optimized XGBoost)	Frequency encoding, risk features, reduced SMOTE, class weights, threshold tuning	High precision + strong recall, deployable	Synthetic dataset; limited temporal data	Extends literature with a balanced & practical model

As shown in Table 5.1, this project extends prior work by combining frequency-based features, risk scoring, and threshold tuning within an optimized XGBoost framework.

This integration results in a balanced model that achieves both high precision and strong recall, addressing limitations identified in previous studies.

5.5 Discussion of Model Behaviour

The behaviour of both baseline and optimized models reflects the inherent challenges of fraud detection: extreme class imbalance, heterogeneous customer behaviour, high-cardinality categorical features, and noise in synthetic transaction data. Understanding how the models responded to these challenges provides valuable insight into their strengths, limitations, and practical applicability.

5.5.1 Baseline Model Behaviour

Baseline models - particularly Logistic Regression, Decision Trees, Random Forests, and basic XGBoost exhibited a clear and consistent behavioural pattern:

High Recall, Low Precision

All baseline models identified nearly all fraudulent cases, producing recall values close to 1.0. However, the models incorrectly labelled a significant number of legitimate transactions as fraud, leading to low precision.

This behaviour emerged due to:

- Severe class imbalance: The minority class (fraud) is extremely small relative to legitimate transactions.
- Oversampling alone is not sufficient: Although SMOTE expanded the fraud class during training, models still tended to generalize fraud-like patterns too broadly.
- Weak boundaries in raw feature space: Without risk-based or engineered features, model decision boundaries remained diffuse and prone to misclassification.

The baseline behaviour thus reflects the difficulty of learning discriminative fraud signals from raw or lightly transformed inputs.

5.5.2 Optimized Model Behaviour

The optimized XGBoost model demonstrated a markedly different behavioural profile from the baselines. This change stems from enhanced feature engineering, improved sampling strategy, and threshold adjustments.

***Substantial Increase in Precision**

Precision rose from approximately 0.20 in the baseline to over 0.77 in the optimized model. This dramatic improvement indicates that:

- The model became far more selective in assigning the fraud label.
- The risk-based features strengthened the model's ability to differentiate legitimate vs. fraudulent behaviour.
- Conservative SMOTE (10%) prevented the model from overgeneralizing fraud-like patterns.

This behavioural shift is essential for real-world fraud detection, where false alarms are costly and disruptive.

***Moderate Decrease in Recall**

The optimized model's recall decreased from nearly perfect detection to around 0.73. This reflects a deliberate trade-off:

- The stricter decision boundary removed ambiguous or borderline cases.
- Threshold tuning prioritised confidence over coverage.
- A reduction in synthetic samples led to tighter clustering of fraud cases.

While this means some fraudulent transactions are missed, the trade-off is justified in many financial contexts where too many false positives can overwhelm human analysts or hinder customer experience.

***Improved F1-score and Balanced Behaviour**

The F1-score representing the balance between precision and recall more than doubled after optimization. This shows that:

The optimized model is not merely performing well in one metric;

It achieves a balanced improvement across multiple behavioural dimensions.

This confirms that feature engineering and sampling refinement created a more structured, consistent decision-making profile.

5.5.3 Behaviour Reflected in the Confusion Matrix

The confusion matrix of the optimized model illustrates its behavioural pattern:

- True Positives (TP): Remain high due to improved feature representativeness.
- False Positives (FP): Decline sharply, indicating more reliable fraud flags.
- False Negatives (FN): Increase slightly, an expected consequence of prioritising precision.
- True Negatives (TN): Increase significantly, showing improved classification of legitimate transactions.

This behaviour is desirable in production environments where the cost of a false alarm is typically higher than the cost of a missed fraud, especially for low-risk transactions.

5.5.4 Behavioural Implications for Real-World Deployment

The behavioural shift of the optimized XGBoost model demonstrates several practical strengths:

- **Reduced operational load:** Fewer false alarms mean fewer unnecessary manual reviews.
- **Improved customer satisfaction:** Legitimate transactions are less likely to be blocked or flagged.
- **High-risk fraud still captured:** Although recall is lower, the model remains effective at detecting the majority of fraud cases.
- **Strong ranking capability (ROC - AUC ≈ 0.997)** ensures stable risk scoring even if thresholds change.

These behavioural characteristics make the optimized model suitable for integration into real-time fraud screening pipelines, where precision-driven models tend to be more acceptable.

Chapter 6: Conclusions and Future Work

6.1 Conclusions

This study developed and evaluated a machine learning - based fraud detection system using the synthetic BankSim dataset. The project aimed to address the challenges posed by extreme class imbalance, evolving fraud patterns, and the need for models that are both accurate and operationally efficient.

The results demonstrated that baseline models - Logistic Regression, Decision Tree, Random Forest, and baseline XGBoost - performed well in terms of recall due to the use of SMOTE, but suffered from extremely low precision. Such models correctly identified almost all fraudulent transactions but produced a large number of false positives, making them impractical for real-world deployment.

To overcome these limitations, an optimized XGBoost model was developed using a combination of frequency encoding, risk-based feature engineering, controlled oversampling, class weighting, hyperparameter tuning, and probability threshold optimization. This model achieved a substantially better trade-off between precision and recall, significantly improving F1-score while maintaining excellent ROC - AUC performance. These enhancements reduced the number of false alarms and made the system more suitable for practical use cases in financial institutions.

Overall, the findings confirm that effective fraud detection requires more than simply choosing a strong classifier. Instead, it depends heavily on data preprocessing, imbalance-handling strategies, engineered behavioral features, and careful threshold tuning. The methodology proposed in this study provides a strong and extensible framework for fraud detection tasks and aligns well with trends found in existing literature.

6.2 Future Work

While the project achieved strong results, several extensions could further enhance performance, interpretability, and applicability to real-world financial systems.

6.2.1 Incorporating Temporal and Sequential Modelling

Future research should expand beyond static features to include:

- inter-transaction time intervals,
- customer behavioural trends,
- time-of-day and periodic patterns,
- session-based modelling.

Models such as LSTM, GRU, or Transformer architectures could better capture evolving fraud patterns.

6.2.2 Graph-Based Fraud Modelling

Fraud often spreads through complex networks involving customers, merchants, devices, and locations.

Graph-based learning approaches such as Graph Neural Networks (GNNs) can uncover relational anomalies that traditional tabular models may miss. This is a promising direction for detecting collusive or organized fraud schemes.

6.2.3 Integration of Real-World Banking Data

Validating the model on real, anonymized financial data would strengthen the external validity of the findings. Real transaction logs contain:

- richer behavioural patterns,
- a wider diversity of fraudulent behaviours,
- complex temporal dependencies.

Testing on real data would also enable cost-sensitive evaluation, which is essential for financial decision-making.

6.2.4 Advanced Hyperparameter Optimization

Although RandomizedSearchCV improved performance, further optimization could be conducted using:

- Bayesian optimization,
- genetic algorithms,
- bandit-based hyperparameter tuning,
- dynamic threshold adjustment techniques.

These approaches may further increase precision and recall while minimizing computational costs.

6.2.5 Enhancing Explainability and Trust

To improve adoption in regulated industries, future work could focus on:

- integrating SHAP-based interpretability dashboards,
- combining rule-based logic with machine learning outputs,
- producing human-readable explanations for fraud alerts.

Explainable AI (XAI) methods would help ensure transparency in automated fraud detection systems.

6.2.6 Real-Time Deployment and System Integration

Future extensions may explore:

- latency optimization for real-time scoring,
- streaming data processing (Apache Kafka, Spark Streaming),
- continuous learning systems that adapt to new fraud patterns,
- A/B testing with live transaction pipelines.

Real-world deployment requires not only a strong model but also engineering considerations for scalability and monitoring.

6.3 Final Remarks

This study demonstrates that machine learning - particularly ensemble boosting methods - provides a compelling approach to fraud detection. By combining rigorous data preprocessing, sophisticated feature engineering, and targeted optimization, the resulting model achieves strong performance while maintaining practical operational value.

Although challenges remain, especially regarding temporal modelling, dataset realism, and interpretability, the project establishes a robust foundation for building scalable and accurate fraud detection systems. With further refinement and integration into real-world financial

infrastructure, such models have the potential to significantly reduce fraud losses and improve customer trust.

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